NYC Taxi Trip Data – Exploratory Analysis Summary

Executive summary report

Prepared by Priyanka Rajangale | Google Advanced Data Analytics Certificate

OVERVIEW

This project analyzes 22,699 NYC taxi trip records containing 18 variables using Python and Pandas. The objective is to understand trip characteristics, identify unusual fare values, examine payment patterns, and determine important variables for future predictive modeling. The findings will support data-driven decision-making for the New York City Taxi and Limousine Commission (NYC TLC).

PROJECT STATUS

- Inspected 22,699 NYC taxi trip records across 18 variables using Python and Pandas.
- Checked data completeness and identified no missing values.
- Examined trip distances, fares, payment methods, and vendor records.
- Identified unusual negative and zero fare amounts requiring further investigation.
- Analyzed average tips and vendor-level charges to support future predictive modeling.

NEXT STEPS

1. Investigate negative and zero fare amounts to identify possible data errors or adjustments.
2. Examine unusually long trips and potential outliers.
3. Convert pickup and drop-off timestamps into datetime format for further analysis.
4. Explore relationships between trip distance, total fare, and other variables before building predictive models.

KEY INSIGHTS

- **Trip Distance:** Average 2.91 miles; median 1.61 miles; maximum 33.96 miles, indicating a right-skewed distribution.
- **Fare Anomalies:** Negative and zero charges were identified, with a minimum total amount of -$120.30.
- **Payment Methods:** 15,265 credit card trips and 7,267 cash trips were recorded.
- **Tips:** Average recorded credit card tip was $2.73, compared with $0.00 for cash payments.
- **Vendor Comparison:** Average total charges were $16.30 for Vendor 1 and $16.32 for Vendor 2.

NYC Taxi Trip Data – Summary Statistics

Metric	Value
Total taxi trips	22,699
Average trip distance	2.91 miles
Maximum trip distance	33.96 miles
Credit card trips	15,265
Cash trips	7,267
Average credit card tip	$2.73
Minimum total charge	-$120.30